

Plenum Box

Plenum boxes are used to collect air before it enters the outlets directly. This reduces the air velocity, absorbs flow changes, reduces noise, and ensures uniform air distribution along the outlets.

Box Plenum Models:

The shapes of the boxes vary depending on the shape of the outlets. The following are the most important box shapes:

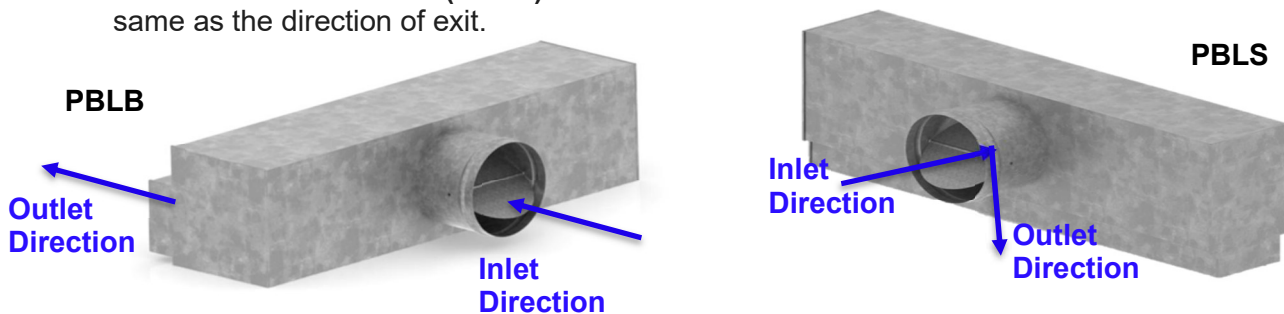
- **Straight plenum boxes:** Ideal for applications involving linear outlets like LSD, FBD, LBG or even AGs.
- **Curved plenum boxes:** Ideal for applications involving curved outlets like curved (LSD, FBD, LBG, AG).
- **Square/Rectangular Plenum boxes (PBSD):** Ideal for installation on square or rectangular diffusers for ceiling use only. Air entry can be from the top or from the side.



Straight PB Models:

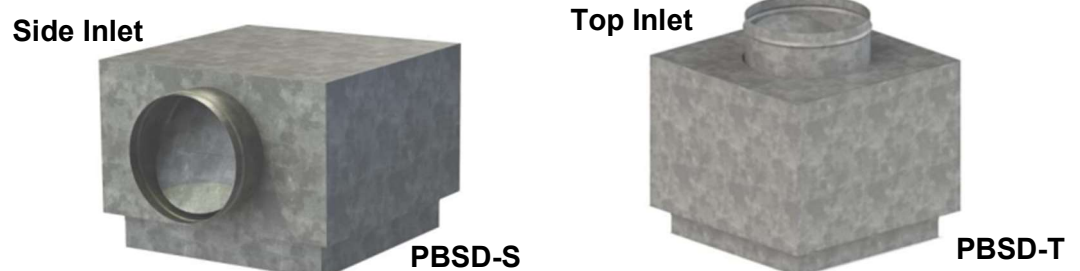
The shapes of the boxes vary depending on the direction of air flow inside the box.

- **Plenum Box Linear Slot (PBLB):** In this model the air movement inside, it is in the shape of the letter L, meaning that the direction of air exit is perpendicular to the direction of entry.
- **Plenum Box Linear Bar (PBLB):** In this model the air direction is the same as the direction of exit.



Square PB Models:

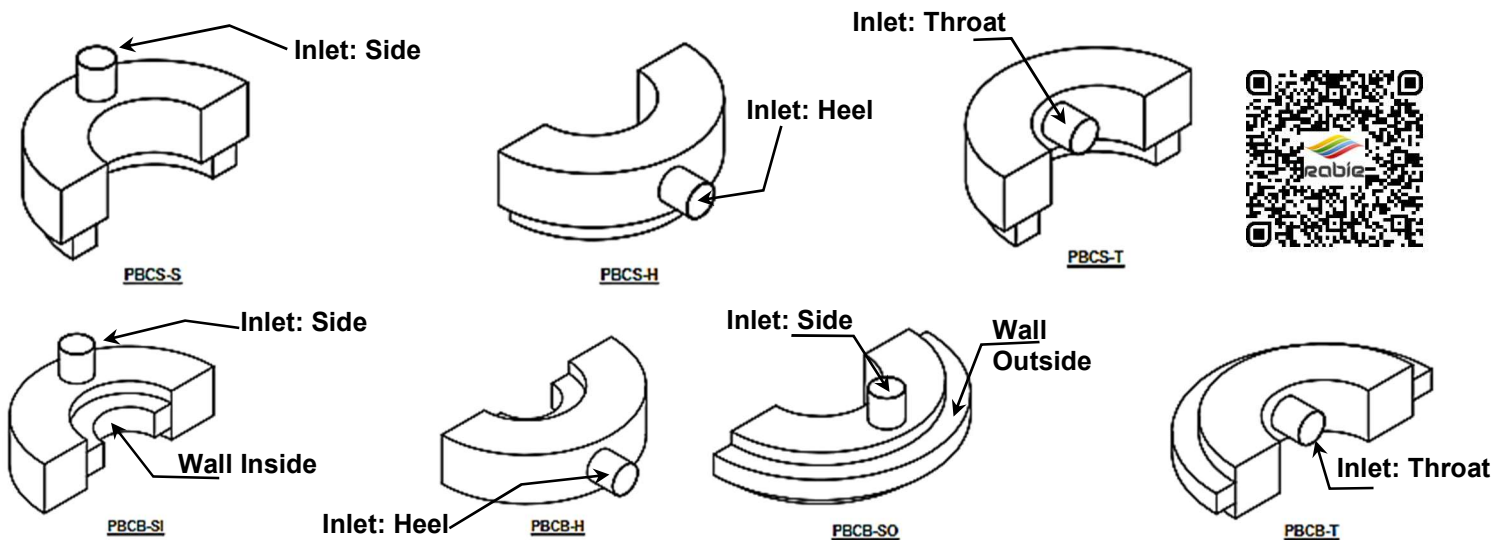
This is used for installation on square or rectangular diffusers for ceiling use only. Air entry can be from the top or from the side.



Curved PB Models:

For curved plenum boxes, the selection can be tricky compared to straight ones, because air flow direction isn't the only factor to consider - there are several! To ensure a perfect fit, we recommend letting your customer choose the ideal curved plenum box model suitable for his needs or by simply describing the suitable type for his needs. Simply by looking through the available model photos and pick the one that matches his requirements.

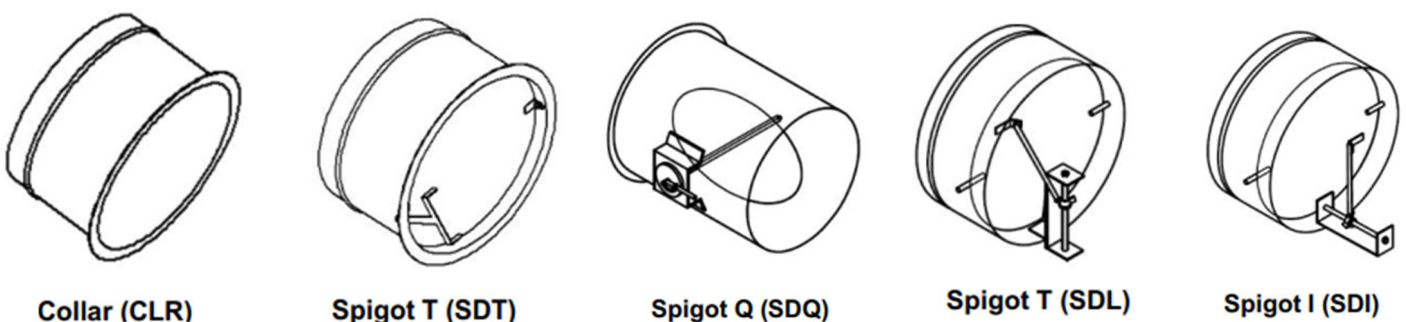
- **Plenum Box Type:** Curved Bar (PBCB) or Curved Slot (PBCS), Noting that PB type is specified here depending on the outlet installation mounting location (PBCS for Ceiling & PBCB for wall).
- **The inlet Location:** Installed from the Heel side or from the Throat side or from the top depending on the customer's request.
- After the customer describes the shape of the box, the representative determines the type of box required based on this description using the first code guide (guide-1) approved by the factory. you can download the model guide using the QR code provided.



Selection Parameters:

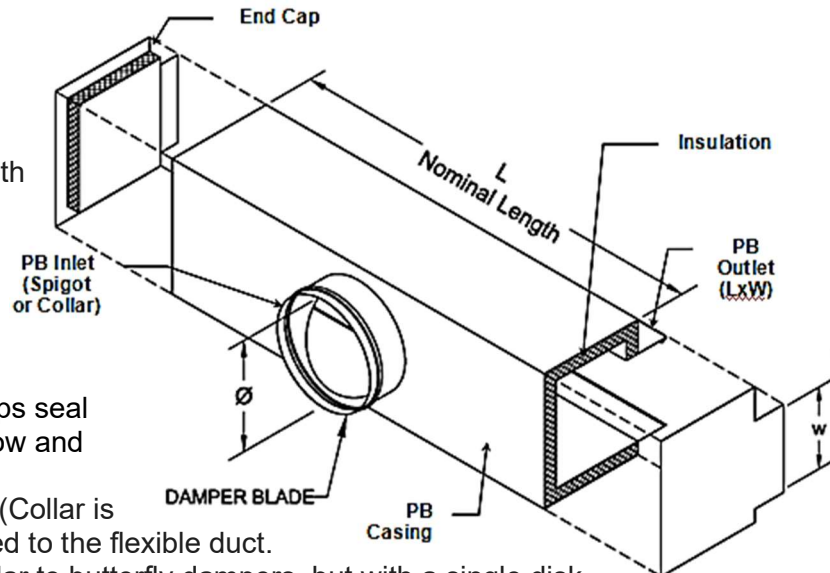
The selection of Plenum boxes is based on defining the following parameters:

- **Model:** Defining the specific PB model as specified above (PBLB, PBLS, PBSD,etc).
- **Outlet:** Specify the desired outlet type for your PB installation (LBG, LSD, SCD, etc.) along with its corresponding size dimensions (LxW or WxH).
- **Inlet:** Defining the Inlet configuration (required):
 - ♦ **Size & Quantity:** Specify the inlet size diameter and their quantity along the PB length.
 - ♦ **Location:** Indicate the desired location on the PB.
 - ♦ **Type:** Choose from available options like collar, spigot tee, etc.



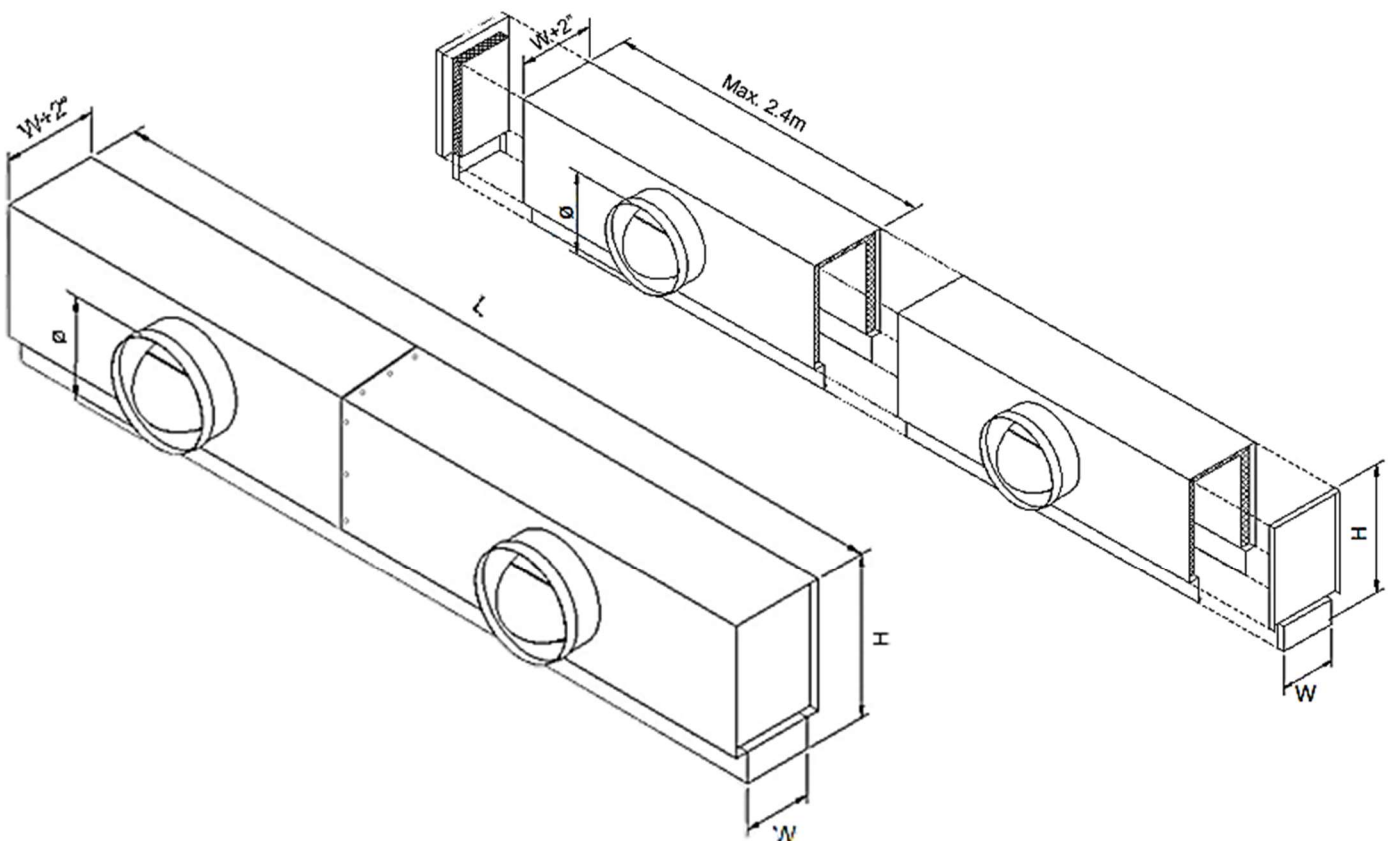
Design Parameters:

- Casing:** The casing is manufactured with a thickness of 0.5 mm and in a shape that matches the outlet shape that will be mounted on it. The box size is requested by specifying the outlet size. The box can be ordered with hanging angles to facilitate installation on site.
- End Cap:** Linear & Curved PBs end caps seal unused openings, ensuring proper airflow and structural integrity.
- Inlets:** The inlets are circular openings (Collar is standard) fixed to the box and connected to the flexible duct. They can also be spigot-controlled similar to butterfly dampers, but with a single disk that has a clearance to rotate around its diameter when the screw is turned with a screwdriver. The customer must specify the diameter and number of inlets.
- Insulation:** The box is lined internally using glass fiber insulation with a density of 24 or 48 kg/m³ depending on the request and a thickness of 1 inch. It is fixed using glue or fixing clamps.
- Options:** The plenum box can be customized according to customer needs as follow:
 - Thickness:** we have may options for PB casing thickness (0.6mm, 0.8mm, 1mm,...).
 - Insulation:** The insulation can be Fiber glass 24 or 48 kg/m³ 1" or 2" thick or it can be rubber with (0.25mm, 0.5mm, 1mm thick).
 - End Cap:** End cap can be outside air stream.



Multi Section Plenum Box:

Due to limitations in material handling, Transportation and fabrication, linear and curved plenum boxes exceeding 2.4 meters in length are constructed as multi-section, with a maximum individual section length of 2.4 meters. These sections are subsequently assembled on-site and secured with an end cap to achieve the specified total length.

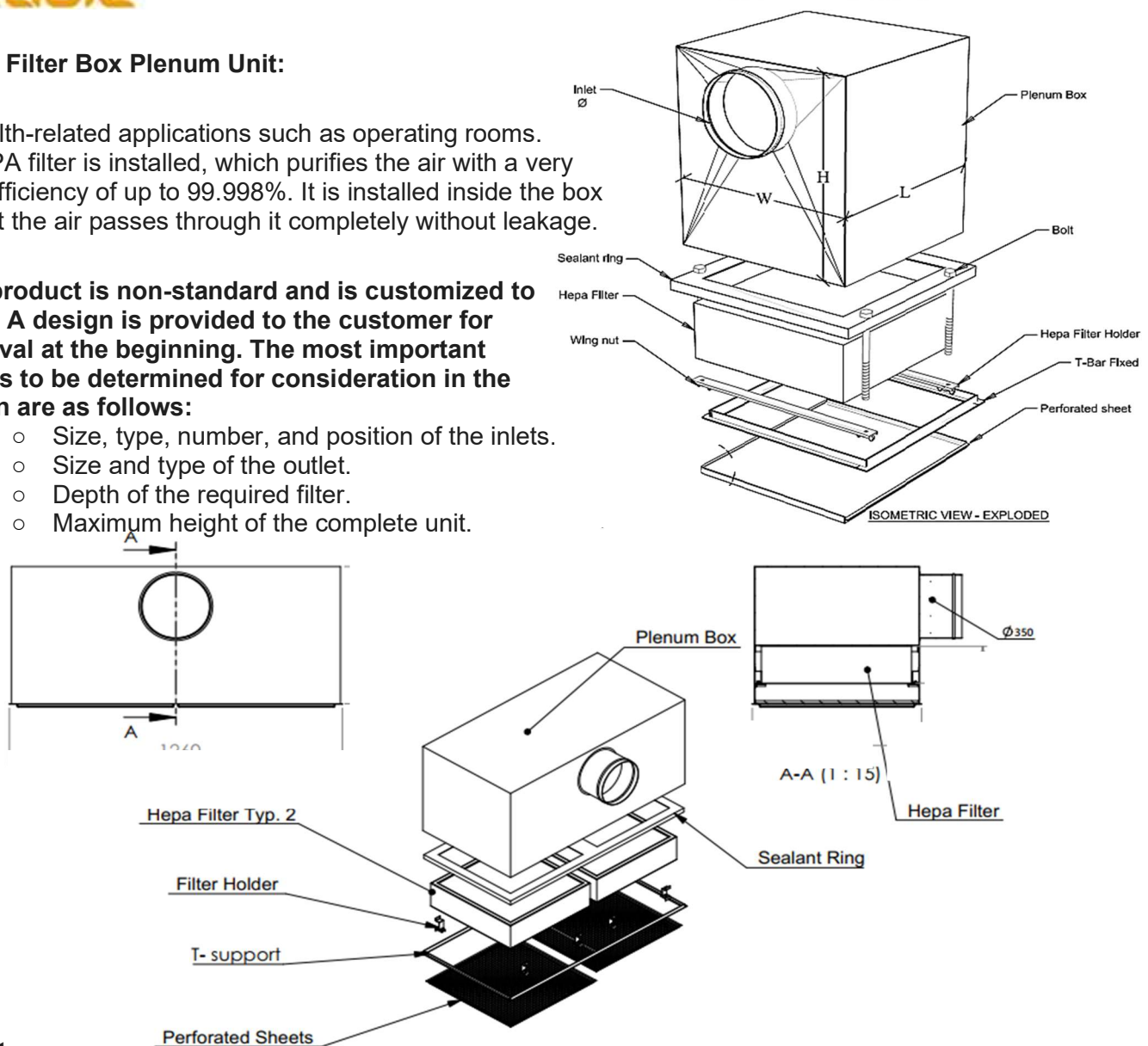


HEPA Filter Box Plenum Unit:

In health-related applications such as operating rooms. A HEPA filter is installed, which purifies the air with a very high efficiency of up to 99.998%. It is installed inside the box so that the air passes through it completely without leakage.

This product is non-standard and is customized to order. A design is provided to the customer for approval at the beginning. The most important factors to be determined for consideration in the design are as follows:

- Size, type, number, and position of the inlets.
- Size and type of the outlet.
- Depth of the required filter.
- Maximum height of the complete unit.

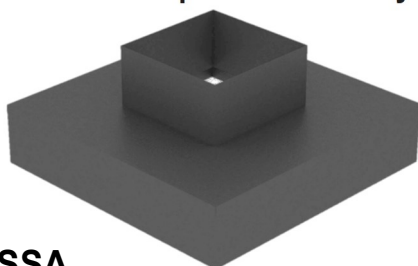


Adapter:

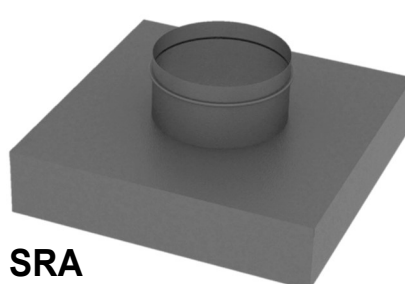
An adapter represents the simplest form of a plenum box without insulation and in the smallest size. It is economical. Its generally used to connect 2 pieced of different sizes or different geometry, It can be with sharp transition or smooth Transition.

There are two standard types of adapters:

- **Round to Square Adapter:** This is the most common type and is typically used for installation on square diffusers.
- **Square to Square Adapter:** The adapter is painted black to be concealed.
- **The customer specifies the inlet and outlet size only, and can also specify the height of the adapter if necessary.**



SSA
SQUARE TO SQUARE ADAPTER



SRA
SQUARE TO ROUND ADAPTER



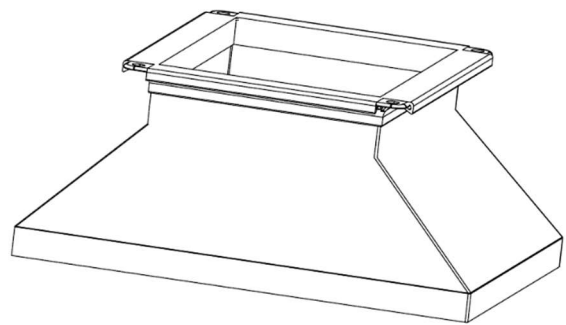
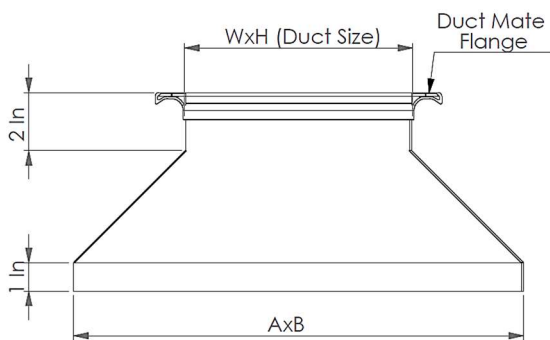
SRA-T
SQUARE TO ROUND ADAPTER
WITH TRANSITION (SMOOTH)

Bell Mouth (BM)

Bell Mouth is primarily designed and used for return or extract air purposes, more commonly located within ceiling voids. The bell mouth cross-sectional area is normally double that of the duct area so that the air velocity entering the bell mouth is low (to reduce noise, turbulence and pressure drop), and gradually increases to the normal design velocity of the ductwork. The angle of the bell mouth is normally tapered at about 45° as a balance between keeping the bell mouth short without causing too much turbulence or excessive pressure drop.

Bell Mouth is non-standard and is customized to each order. The most important factors to be considered while ordering the BM are as follows:

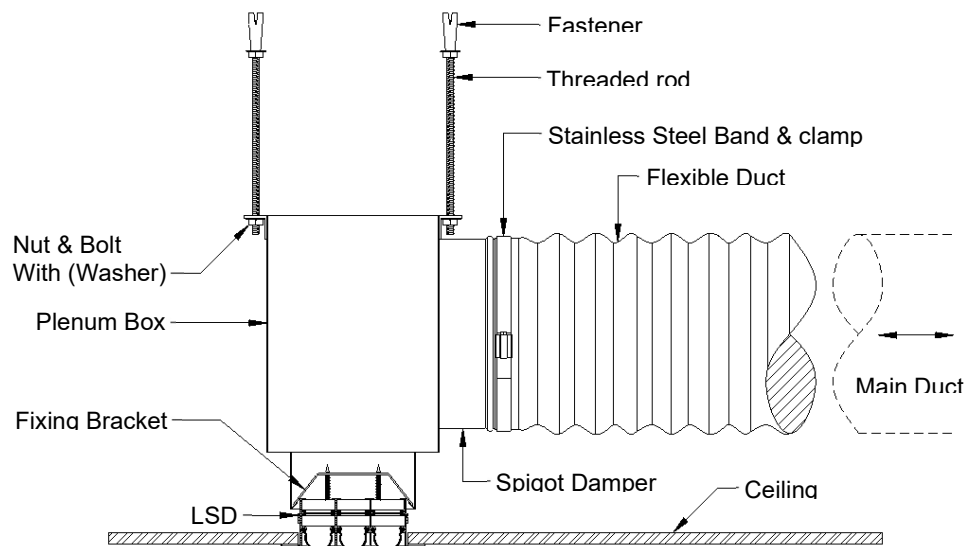
- Size (Both for Inlet & Outlet).
- Inlet End Type (Flanged, Duct mate flange (Std.),...other.
- Screen Type (Without, Bird Screen, Insect Screen).
- Optional dimensions requirements (Total depth, Inclination angle, Inclined at 1 side only,...)



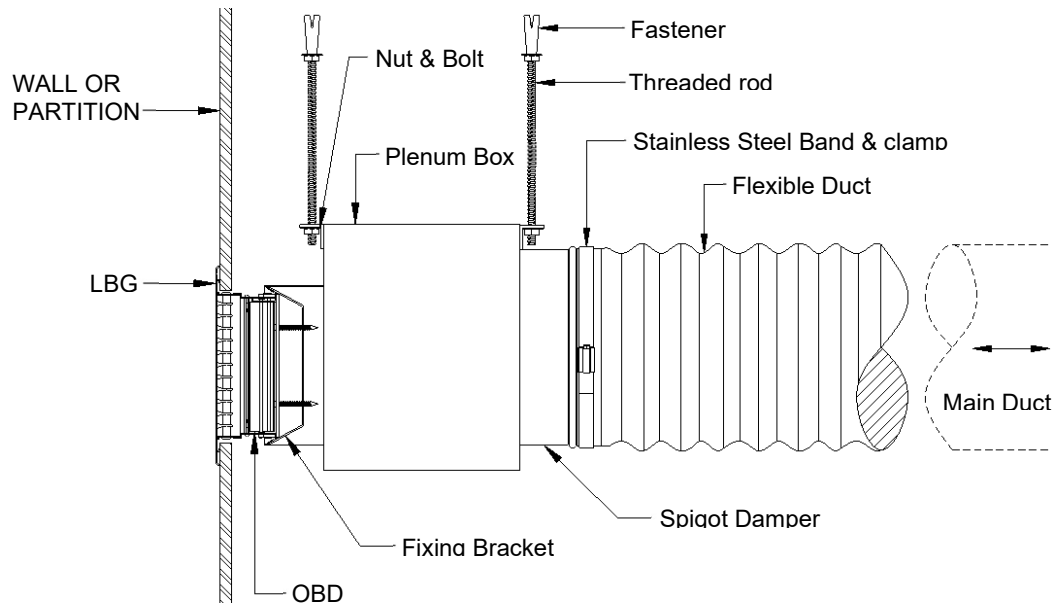
NOTE: Optional wire mesh can be added at the larger side. It can be bird or insect type, and the material can be GI (Standard) or aluminum.

Installation

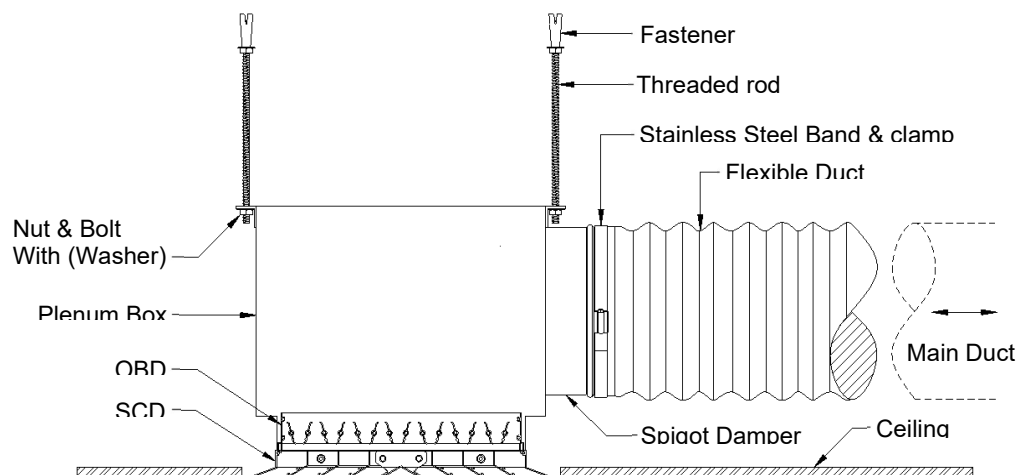
Installation: LSD (PBLs)



Installation: LBG (PBLB)



Installation: SCD (PBSD)



Duct Access Doors

Duct access doors are used to provide access to the interior of air ducts for inspection and maintenance purposes. For example, to replace a fire damper fusible link. They are made of galvanized steel and have a panel that is attached to a frame. The panel can be opened in one of two ways: with a cam-hinge or with two cams.

Components:

Duct access doors are manufactured from galvanized steel sheets with a thickness of 0.5 mm and the following components:

- **Frame:** The frame is cut to the dimensions of the duct opening. It is usually a U-shaped frame with a neck that is inserted into the duct opening and then bent around the opening with the neck bent in a zigzag pattern to facilitate installation.
- **Panel:** The panel is a sandwich of steel filled with fiberglass insulation. It is rectangular in shape and 1 inch thick. It is attached to the frame in one of two ways depending on the model.
- **Cam-hinge door (DAD-CH):** The panel is attached to the frame with hinges on one side and a cam on the other side. The panel opens and closes by rotating around the hinge like a conventional door.
- **Cam-cam door (DAD-CC):** The panel is attached to the frame on both sides using a cam on each side. This allows the panel to be removed completely when the door is opened.
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Additional information:

- More than one cam can be placed on each side depending on the size of the door, the manufacturer's standard specifications, or the customer's request. The hinge or cam can also be placed on the larger or smaller side of the door.
- If the customer does not specify the distribution of the cams around the door, they will be distributed according to the manufacturer's specifications.
- The direction of opening is considered to be outward from the duct if not otherwise specified.
- A rubber gasket is attached around the perimeter of the joint between the panel and the frame to prevent air leakage when the door is closed.